SAFETY DATA SHEET

CONCLUSION 300 SC

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	2025/12/03	50003108	Date of first issue: 2025/12/03

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : CONCLUSION 300 SC

Recommended use of the chemical and restrictions on use

Recommended use : Insecticide

Restrictions on use : Use as recommended by the label.

Manufacturer or supplier's details

Company : FMC Agro Philippines, Inc.

Address : 32nd Floor Seven Neo Building,
5th Avenue between 26th and 28th Street,
Bonifacio Global City, Taguig City 1634
Philippines

Telephone : +63 (2) 5323-5750

E-mail address : SDS-Info@fmc.com

National Poison Control Center : U.P. PGH, Padre Faura, Manila (+63) 2 8524 1078
East Avenue, Quezon City (+63) 2 8928 0611
Southern Philippines Medical Center (+63) 82 227 2731
(formerly Davao Medical Center Davao City)

Emergency telephone : For leak, fire, spill or accident emergencies, call:
+(63) 2-395-3308 (CHEMTREC)
Toll-free mobile enabled: 1800 1 322 0553 (CHEMTREC)

Medical emergency:
All other countries: +1 651 / 632-6793 (Collect)

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 4

Specific target organ toxicity - single exposure : Category 2 (Central nervous system)

Specific target organ toxicity - repeated exposure : Category 2 (Central nervous system)

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS label elements

Hazard pictograms :

Signal Word : WARNING

Hazard Statements :
H302 Harmful if swallowed.
H371 May cause damage to organs (Central nervous system).
H373 May cause damage to organs (Central nervous system) through prolonged or repeated exposure.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P260 Do not breathe mist or vapors.
P264 Wash skin thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P273 Avoid release to the environment.

Response:

P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER/ doctor if you feel unwell. Rinse mouth.
P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.
P391 Collect spillage.

Storage:

P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
imidacloprid (ISO)	138261-41-3	20 -25
Bifenthrin	82657-04-3	3 -6
Sodium alkylnaphthalenesulfonate, formaldehyde condensate	68425-94-5	1 -3

4. FIRST AID MEASURES

General advice	:	Move out of dangerous area. Show this material safety data sheet to the doctor in attendance. Do not leave the victim unattended.
If inhaled	:	If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a physician.
In case of skin contact	:	If skin irritation persists, call a physician. If on skin, rinse well with water. If on clothes, remove clothes.
In case of eye contact	:	Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Induce vomiting immediately and call a physician. Keep respiratory tract clear. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician. Take victim immediately to hospital.
Most important symptoms and effects, both acute and delayed	:	Harmful if swallowed. May cause damage to organs. May cause damage to organs through prolonged or repeated exposure.
Notes to physician	:	Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable extinguishing media	:	Carbon dioxide (CO_2)
Unsuitable extinguishing media	:	High volume water jet Do not spread spilled material with high-pressure water streams.
Specific hazards during fire fighting	:	Do not allow run-off from fire fighting to enter drains or water courses.
Hazardous combustion products	:	Fire may produce irritating, corrosive and/or toxic gases. Chlorinated compounds Nitrogen oxides (NOx) Carbon oxides Hydrogen cyanide Fluorinated compounds

Specific extinguishing methods	:	Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. For safety reasons in case of fire, cans should be stored separately in closed containments. Use a water spray to cool fully closed containers.
Special protective equipment for fire-fighters	:	Firefighters should wear protective clothing and self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	:	Use personal protective equipment. Keep people away from and upwind of spill/leak.
Environmental precautions	:	Prevent product from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	:	Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	:	Do not spray on a naked flame or any incandescent material. Keep away from open flames, hot surfaces and sources of ignition.
Advice on safe handling	:	Avoid formation of aerosol. Do not breathe vapors/dust. Avoid contact with skin and eyes. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area. Provide sufficient air exchange and/or exhaust in work rooms. Dispose of rinse water in accordance with local and national regulations.
Conditions for safe storage	:	No smoking. Keep in a well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability	:	No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

Respiratory protection	:	In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.
Hand protection		
Material	:	Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.
Remarks	:	The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	:	Eye wash bottle with pure water Tightly fitting safety goggles
Skin and body protection	:	Impervious clothing Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Protective measures	:	Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions.
Hygiene measures	:	When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Form	:	liquid
Color	:	white
Odor	:	No data available
pH	:	5 - 8 (25 °C) Method: CIPAC MT 75
Melting point/freezing point	:	No data available

Boiling point/boiling range	:	No data available
Flash point	:	> 77 °C
Flammability (liquids)	:	Not expected to be ignitable
Self-ignition	:	No data available
Relative density	:	1.08 - 1.14
Partition coefficient: n-octanol/water	:	Not applicable
Viscosity		
Viscosity, kinematic	:	516 mm2/s (24 °C) Method: OECD Test Guideline 114
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Particle size	:	Not applicable

10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed. Vapors may form explosive mixture with air.
Conditions to avoid	:	Heat, flames and sparks.
Incompatible materials	:	Strong acids and strong bases Strong oxidizing agents
Hazardous decomposition products	:	No hazardous decomposition products are known.

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Harmful if swallowed.

Product:

Acute oral toxicity	:	LD50 (Rat): 1,098 mg/kg Method: OECD Test Guideline 425
Acute inhalation toxicity	:	LC50 (Rat): > 2.05 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Assessment: The substance or mixture has no acute inhalation toxicity Remarks: Highest attainable concentration.
Acute dermal toxicity	:	LD50 (Rat): > 4,000 mg/kg Method: OECD Test Guideline 402

Components:

imidacloprid (ISO):

Acute oral toxicity	:	LD50 (Rat, male and female): > 1,000 mg/kg Symptoms: Tremors, piloerection, Breathing difficulties Remarks: no mortality LD50 (Rat, female): 300 - 2,000 mg/kg Method: OECD Test Guideline 423 Symptoms: Fatality, Convulsions, piloerection GLP: yes Assessment: The component/mixture is moderately toxic after single ingestion. LD50 (Rat, female): 300 - 2,000 mg/kg Method: OECD Test Guideline 420 Symptoms: Fatality, Tremors, ataxia GLP: yes Assessment: The component/mixture is moderately toxic after single ingestion. LD50 (Rat, female): ca. 2,567 mg/kg Method: OECD Test Guideline 425 Symptoms: Fatality, Breathing difficulties GLP: yes
Acute inhalation toxicity	:	LC50 (Rat, male and female): > 5.31 mg/l Exposure time: 4 h Test atmosphere: dust/mist Assessment: The substance or mixture has no acute inhalation toxicity Remarks: no mortality LC50 (Rat, male and female): 5.17 mg/l Exposure time: 4 h Method: OECD Test Guideline 403 Symptoms: hypoactivity GLP: yes

		Assessment: The substance or mixture has no acute inhalation toxicity Remarks: no mortality
		LC50 (Rat, male and female): > 4.9 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: Breathing difficulties, ataxia, Convulsions, Tremors Assessment: The component/mixture is minimally toxic after short term inhalation.
Acute dermal toxicity	:	LD50 (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 402 Symptoms: Irritation GLP: yes Assessment: The substance or mixture has no acute dermal toxicity Remarks: no mortality
		LD50 (Rabbit): > 2,000 mg/kg

Bifenthrin:

Acute oral toxicity	:	LD50 (Rat, male and female): 56.7 mg/kg Symptoms: Convulsions, Tremors, ataxia
		LD50 (Mouse, female): 42.5 mg/kg Method: OPPTS 870.1100
Acute inhalation toxicity	:	LC50 (Rat, female): 0.6 - 1.2 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: Tremors, Convulsions
		LC50 (Rat, male): 1.10 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: Tremors, Fatality
Acute dermal toxicity	:	LD50 (Rat, male and female): > 2,000 mg/kg Remarks: no mortality

Sodium alkylnaphthalenesulfonate, formaldehyde condensate:

Acute oral toxicity	:	LD50 (Rat): > 5,000 mg/kg

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation

Components:

imidacloprid (ISO):

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	No skin irritation
GLP	:	yes

Bifenthrin:

Species	:	Rabbit
Method	:	OECD Test Guideline 404
Result	:	slight or no skin irritation.
GLP	:	yes

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405

Components:

imidacloprid (ISO):

Species	:	Rabbit
Result	:	No eye irritation
Method	:	OECD Test Guideline 405
GLP	:	yes

Bifenthrin:

Species	:	Rabbit
Result	:	Slight or no eye irritation
Method	:	OECD Test Guideline 405
GLP	:	yes

Sodium alkylnaphthalenesulfonate, formaldehyde condensate:

Result	:	Eye irritation

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Routes of exposure	:	Skin contact
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.

Components:

imidacloprid (ISO):

Test Type	:	Maximization Test
Species	:	Guinea pig
Result	:	Does not cause skin sensitization.
Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Method	:	OECD Test Guideline 429
Result	:	Does not cause skin sensitization.
GLP	:	yes

Bifenthrin:

Test Type	:	Maximization Test
Routes of exposure	:	Skin contact
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.
GLP	:	yes

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Test system: Chinese hamster cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)

Result: negative
GLP: yes

Genotoxicity in vivo : Test Type: Cytogenetic assay
Species: Chinese hamster
Result: negative
GLP: yes

Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative
GLP: yes

Test Type: dominant lethal test
Species: Mouse
Result: negative

Test Type: chromosome aberration assay
Species: Mouse
Result: negative

Bifenthrin:

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: gene mutation test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes

Test Type: Mouse lymphoma assay
Metabolic activation: with and without metabolic activation
Result: negative

Genotoxicity in vivo : Test Type: Sex-linked Recessive Lethal Test
Species: Drosophila melanogaster (vinegar fly)
Result: negative

Test Type: unscheduled DNA synthesis assay
Species: Rat
Method: OECD Test Guideline 486
Result: negative

Test Type: Mammalian erythrocyte micronucleus test (in vivo cytogenetic assay)

Species: Mouse (male and female)
Method: OECD Test Guideline 474
Result: negative

Carcinogenicity

Based on available data, the classification criteria are not met.

Components:

Bifenthrin:

Species	:	Rat, female
Application Route	:	Oral
Exposure time	:	2 Years
NOAEL	:	3 mg/kg bw/day
Result	:	negative
Species	:	Mouse, male
Application Route	:	Oral
Exposure time	:	18 month(s)
NOAEL	:	7.6 mg/kg bw/day
Result	:	positive
Symptoms	:	malignant tumors

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

Effects on fertility	:	Method: OECD Test Guideline 416 Result: Animal testing did not show any effects on fertility. Method: OECD Test Guideline 416 Result: No effects on fertility and early embryonic development were detected.
Effects on fetal development	:	Species: Rabbit Application Route: Oral Dose: 0, 8, 24, 72 mg/kg bw/day General Toxicity Maternal: NOAEL: 8 mg/kg bw/day Method: OECD Test Guideline 414 Result: No teratogenic effects. GLP: yes Species: Rat Dose: 0, 10, 30, 100 mg/kg bw/day General Toxicity Maternal: NOEL: 10 mg/kg bw/day Embryo-fetal toxicity.: NOEL: 30 mg/kg bw/day Method: OECD Test Guideline 414 GLP: yes Test Type: Multi-generation study Species: Rat

Application Route: Oral
Dose: 8, 20, 56 mg/kg bw/day
General Toxicity Maternal: NOEL: 20 mg/kg body weight
Developmental Toxicity: NOEL: 20 mg/kg body weight
Result: No teratogenic effects.
GLP: yes

Bifenthrin:

Effects on fertility : Test Type: Two-generation study
Species: Rat
Application Route: Oral
General Toxicity Parent: NOAEL: 3 mg/kg bw/day
General Toxicity F1: NOAEL: 5 mg/kg bw/day
Result: negative

Effects on fetal development : Test Type: Embryo-fetal development
Species: Rabbit
Application Route: Oral
General Toxicity Maternal: NOAEL: 2.7 mg/kg bw/day
Teratogenicity: NOAEL: 2.7 mg/kg bw/day
Symptoms: Maternal effects.
Result: No teratogenic effects.

Test Type: Embryo-fetal development
Species: Rat
Application Route: Oral
General Toxicity Maternal: NOAEL: 1 mg/kg bw/day
Teratogenicity: NOAEL: 2 mg/kg bw/day
Result: No teratogenic effects.

Species: Rat
Application Route: Oral
General Toxicity Maternal: LOAEL: 7.2 mg/kg bw/day
Developmental Toxicity: LOAEL: 7.2 mg/kg bw/day
Embryo-fetal toxicity.: NOEL: 9.0 mg/kg bw/day
Method: OECD Test Guideline 426
Result: Animal testing did not show any effects on fertility., Some evidence of adverse effects on development, based on animal experiments.

STOT-single exposure

May cause damage to organs (Central nervous system).

Components:

Bifenthrin:

Target Organs : Central nervous system
Assessment : Causes damage to organs.

STOT-repeated exposure

May cause damage to organs (Central nervous system) through prolonged or repeated exposure.

Components:

Bifenthrin:

Target Organs	:	Central nervous system
Assessment	:	The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Repeated dose toxicity

Components:

imidacloprid (ISO):

Species	:	Dog
NOEL	:	1200 ppm
Application Route	:	Oral - feed
Exposure time	:	90 d
Method	:	OECD Test Guideline 409
GLP	:	yes

Species	:	Dog
LOAEL	:	49 mg/kg
Application Route	:	Oral - feed
Exposure time	:	28 d
Dose	:	0, 7.3, 31, 49 mg/kg bw/day
Method	:	OECD Test Guideline 409
Symptoms	:	Tremors, ataxia, Vomiting

Species	:	Dog, male and female
NOEL	:	72 mg/kg bw/day
Application Route	:	Oral - feed
Exposure time	:	52 w
Dose	:	0, 6.1, 15, 41, 72 mg/kg bw/day
GLP	:	yes

Bifenthrin:

Species	:	Rat, male and female
NOEL	:	100 ppm
Application Route	:	Oral - feed
Exposure time	:	90 d
Remarks	:	No toxicologically significant effects were found.

Species	:	Dog, male and female
NOEL	:	2.5 mg/kg bw/day
Application Route	:	Oral - feed
Exposure time	:	13 w
Symptoms	:	Tremors

Species	:	Dog, male and female
NOEL	:	1.5 mg/kg
Application Route	:	Oral
Exposure time	:	52 w
Dose	:	0.75, 1.5, 3, 5 mg/kg bw/day
GLP	:	yes

Symptoms : Tremors

Aspiration toxicity

Based on available data, the classification criteria are not met.

Components:

imidacloprid (ISO):

The substance does not have properties associated with aspiration hazard potential.

Bifenthrin:

The substance does not have properties associated with aspiration hazard potential.

Further information

Product:

Remarks : No data available

Components:

imidacloprid (ISO):

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to fish : LC50 (Poecilia reticulata (guppy)): 0.43 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0.38 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Selenastrum capricornutum (green algae)): 0.52 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Components:

imidacloprid (ISO):

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): > 105 mg/l
Exposure time: 96 h
Test Type: static test
Method: EPA OPP 72-1
GLP: yes

LC50 (Salmo gairdneri): 158 - 281 mg/l

		Exposure time: 96 h Test Type: static test Method: OECD Test Guideline 203 GLP: yes
		LC50 (Oncorhynchus mykiss (rainbow trout)): > 83 mg/l Exposure time: 96 h Test Type: static test Method: EPA OPP 72-1 GLP: yes
		LC50 (Cyprinodon variegatus (sheepshead minnow)): 161 mg/l Exposure time: 96 h Test Type: static test GLP: yes
		LC50 (Leuciscus idus (Golden orfe)): 178 - 316 mg/l Exposure time: 96 h Test Type: static test GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 85 mg/l Exposure time: 48 h Method: US EPA Test Guideline OPP 72-2 GLP: yes
		EC50 (Americamysis bahia (mysid shrimp)): 0.0341 mg/l Exposure time: 96 h Test Type: flow-through test Method: US EPA Test Guideline OPP 72-3 GLP: yes
		LC50 (Hyalella azteca (Amphipod)): 0.526 mg/l Exposure time: 96 h Method: US EPA Test Guideline OPP 72-2 GLP: yes
		NOEC (Crassostrea virginica (atlantic oyster)): 23.3 mg/l Exposure time: 96 h Method: US EPA Test Guideline OPP 72-3 GLP: yes
		EC50 (Hyalella azteca (Amphipod)): 0.055 mg/l End point: Immobilization Exposure time: 48 h Method: EPA OPP 72-2
Toxicity to algae/aquatic plants	:	EbC50 (Scenedesmus subspicatus): > 10 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 GLP: yes
		NOEC (Scenedesmus capricornutum (fresh water algae)): >

		119 mg/l Exposure time: 5 d Method: US EPA Test Guideline OPP 122-2 & 123-2
Toxicity to fish (Chronic toxicity)	:	NOEC (Salmo gairdneri): 28.5 mg/l Exposure time: 21 d Method: OECD Test Guideline 204 GLP: yes NOEC (Oncorhynchus mykiss (rainbow trout)): 9.8 mg/l End point: Growth Exposure time: 98 d Test Type: Early Life-Stage Method: US EPA Test Guideline OPP 72-4 GLP: yes NOEC (Oncorhynchus mykiss (rainbow trout)): 9.02 mg/l End point: Hatching success Test Type: flow-through test Method: OECD Test Guideline 210 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 1.8 mg/l Exposure time: 21 d Test Type: semi-static test Method: US EPA Test Guideline OPP 72-4 GLP: yes EC10 (Chironomus riparius (harlequin fly)): 0.00209 mg/l Exposure time: 28 d NOEC (Chironomus tentans): 0.67 µg/l End point: Growth Exposure time: 10 d Test Type: Static renewal test GLP: yes NOEC (Gammarus pulex): 0.064 mg/l End point: Swimming behavior Exposure time: 28 d Test Type: static test Method: OECD 219 GLP: yes
M-Factor (Chronic aquatic toxicity)	:	100
Toxicity to microorganisms	:	IC50 (activated sludge): > 10000
Toxicity to soil dwelling organisms	:	LC50 (Eisenia fetida (earthworms)): 10.7 mg/kg dry weight (d.w.) Exposure time: 14 d
Toxicity to terrestrial organ-	:	LD50 (Coturnix japonica (Japanese quail)): 31 mg/kg

isms

LD50 (Coturnix japonica (Japanese quail)): 2,225 ppm
Exposure time: 5 d

LD50 (Apis mellifera (bees)): 0.0037 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity

LD50 (Apis mellifera (bees)): 0.0081 µg/bee
Exposure time: 48 h

Ecotoxicology Assessment

Other organisms relevant to the environment : Harmful to bees.

Bifenthrin:

Toxicity to fish : LC50 (Salmo gairdneri): 0.00015 mg/l
Exposure time: 96 h
Test Type: flow-through test

LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.00035 mg/l
Exposure time: 96 h
Test Type: flow-through test

LC50 (Oncorhynchus mykiss (rainbow trout)): 0.000256 mg/l
Exposure time: 96 h
Test Type: semi-static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Pimephales promelas (fathead minnow)): 0.000234 mg/l
Exposure time: 96 h
Test Type: semi-static test
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia): 0.00011 mg/l
Exposure time: 48 h

LC50 (Daphnia): 0.0016 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : EC50 (algae): 0.822 mg/l
Exposure time: 72 h

M-Factor (Acute aquatic toxicity) : 1,000

Toxicity to fish (Chronic toxicity) : NOEC (Oncorhynchus mykiss (rainbow trout)): 0.00012 mg/l
Exposure time: 21 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (Daphnia magna (Water flea)): 0.0013 µg/l Exposure time: 21 d
		NOEC (Daphnia magna (Water flea)): 0.00095 µg/l Exposure time: 21 d
M-Factor (Chronic aquatic toxicity)	:	100,000
Toxicity to soil dwelling organisms	:	LD50 (Eisenia fetida (earthworms)): > 16 mg/kg Exposure time: 14 d
Toxicity to terrestrial organisms	:	LD50 (Colinus virginianus (Bobwhite quail)): 1,800 mg/kg
		LD50 (Anas platyrhynchos (Mallard duck)): > 2,150 mg/kg
		LD50 (Apis mellifera (bees)): 0.07 µg a.i./bee Exposure time: 48 h End point: Acute oral toxicity Method: OECD Test Guideline 213
		LD50 (Apis mellifera (bees)): 0.0201 µg a.i./bee Exposure time: 48 h End point: Acute contact toxicity Method: OECD Test Guideline 214
		ED50 (Apis mellifera (bees)): 143 ng ai larva Exposure time: 7 d End point: Acute larval toxicity Method: OECD 237
		ED50 (Apis mellifera (bees)): 15.15 ng ai larva/day Exposure time: 22 d End point: Chronic larval toxicity Method: OECD 239
		NOAEL (Apis mellifera (bees)): 2.30 ng ai larva/day Exposure time: 22 d End point: Chronic larval toxicity Method: OECD 239
		ED50 (Apis mellifera (bees)): 300 ng a.i/bee/day Exposure time: 10 d End point: Chronic oral toxicity Method: OECD TG 245
		NOAEL (Apis mellifera (bees)): 170 ng a.i/bee/day Exposure time: 10 d End point: Chronic oral toxicity Method: OECD TG 245

Sodium alkylnaphthalenesulfonate, formaldehyde condensate:

Toxicity to fish	:	LC50 (Danio rerio (zebra fish)): > 10 - 100 mg/l Exposure time: 96 h Method: OECD Test Guideline 203 Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 100 mg/l Exposure time: 48 h Method: OECD Test Guideline 202 Remarks: Based on data from similar materials
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 Remarks: Based on data from similar materials EC10 (Pseudokirchneriella subcapitata (green algae)): > 100 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 Remarks: Based on data from similar materials
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	EC10 (Daphnia magna (Water flea)): > 10 - 100 mg/l Exposure time: 21 d Method: OECD Test Guideline 211 Remarks: Based on data from similar materials

Persistence and degradability

Components:

imidacloprid (ISO):

Biodegradability	:	Result: Not readily biodegradable.

Bifenthrin:

Biodegradability	:	Result: Not readily biodegradable.
Stability in water	:	Degradation half life (DT50): 2.2 d Hydrolysis: at 60 °C Degradation half life (DT50): 15.6 d Hydrolysis: at 40 °C

Sodium alkylnaphthalenesulfonate, formaldehyde condensate:

Biodegradability	:	Result: Not readily biodegradable. Remarks: Based on data from similar materials

Bioaccumulative potential

Components:

imidacloprid (ISO):

Bioaccumulation	:	Remarks: Low potential for bioaccumulation
Partition coefficient: n-octanol/water	:	log Pow: 0.33 (20 °C) Method: OECD Test Guideline 107

Bifenthrin:

Bioaccumulation	:	Species: Lepomis macrochirus (Bluegill sunfish) Bioconcentration factor (BCF): 1,709 Remarks: Due to the distribution coefficient n-octanol/water, accumulation in organisms is possible.
Partition coefficient: n-octanol/water	:	log Pow: 6.6

Mobility in soil

Components:

imidacloprid (ISO):

Distribution among environmental compartments	:	Koc: 109 - 411 Remarks: Mobile in soils

Bifenthrin:

Distribution among environmental compartments	:	Koc: 236610 ml/g, log Koc: 5.37 Remarks: immobile

Other adverse effects

Product:

Additional ecological information	:	An environmental hazard cannot be excluded in the event of unprofessional handling or disposal. Very toxic to aquatic life with long lasting effects.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	:	The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.
Contaminated packaging	:	Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Bifenthrin, Imidacloprid)
Class	:	9
Packing group	:	III
Labels	:	9
Environmentally hazardous	:	yes

IATA-DGR

UN/ID No.	:	UN 3082
Proper shipping name	:	Environmentally hazardous substance, liquid, n.o.s. (Bifenthrin, Imidacloprid)
Class	:	9
Packing group	:	III
Labels	:	Miscellaneous
Packing instruction (cargo aircraft)	:	964
Packing instruction (passenger aircraft)	:	964
Environmentally hazardous	:	yes

IMDG-Code

UN number	:	UN 3082
Proper shipping name	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Bifenthrin, Imidacloprid)
Class	:	9
Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F
Marine pollutant	:	yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Priority Chemical List (PCL)	:	Not applicable
Chemical Control Order (CCO)	:	Not applicable

The ingredients of this product are reported in the following inventories:

TCSI	:	Not in compliance with the inventory
TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains the following components that are not on the Canadian DSL nor NDSL. imidacloprid (ISO) Bifenthrin Sodium alkylnaphthalenesulfonate, formaldehyde condensate
ENCS	:	Not in compliance with the inventory
ISHL	:	Not in compliance with the inventory
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

16. OTHER INFORMATION

Revision Date	:	2025/12/03
Date format	:	yyyy/mm/dd

Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International

Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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